HOSAKOTA VENKATA NAGA THARUN

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PROFESSIONAL SUMMARY

AI Engineer with 2.5+ years of experience building production generative AI applications on Azure, from document-processing backends and multi-agent assistants to RAG pipelines and AI agents on Workato. Has worked with LLMs and Hugging Face models, and with agent frameworks including LangGraph, Semantic Kernel, and Google ADK. Approaches production issues through root-cause analysis, fixing failures at the source so they do not recur.

EXPERIENCE

Mouri Tech Pvt Ltd **March 2024 - Present**
Applied AI Engineer

Financial Statement Analyzer

- Built a serverless Python backend on Azure Functions with Azure AI Foundry Agents (GPT-4.1, code interpreter) to extract structured summaries from financial statement images; a representative 42-image batch takes ~2 min 50 sec
- Root-caused API rate-limit failures that occurred consistently past 15-20 images per batch under bulk base64 image load, then implemented corrective actions to prevent recurrence: odd/even index routing across two agents, retry logic, and automatic failover between agents; zero rate-limit failures observed since
- Designed a 3-phase prompt pipeline (extraction, verification, summary) with data-fidelity validation to limit hallucination
- Monitored runs through Azure Application Insights logs across monthly and yearly cycles to confirm fixes held
- Secured the service with Azure Identity (DefaultAzureCredential) for passwordless, managed-identity authentication

Multi-Agent AI Assistant Platform

- Designed a multi-agent system with LangGraph StateGraph orchestration, Azure OpenAI (GPT-4.1), and FastAPI; a context-aware router dispatches queries across 6 specialized agents (calculator, weather, HR policy, search, reminder, smalltalk)
- Built a 3-stage hybrid RAG pipeline combining dense FAISS search (70% weight) and sparse BM25 keyword search (30% weight) with cross-encoder reranking, grounding responses across 30+ internal policy documents
- Implemented priority-based short-circuit routing to cut unnecessary LLM calls
- Engineered a human-in-the-loop approval gate for the reminder agent: multi-turn slot filling captures the recipient email, message, and send time, which are presented for explicit user confirmation before any email is dispatched
- Implemented context-aware follow-up handling across agents: recovers the previously invoked tool, determines whether a request continues the prior turn or depends on earlier history (up to 10 turns), and re-triggers approval before repeating an action such as resending a reminder
- Added real-time SSE streaming of graph execution and a dashboard tracking token usage, latency, and agent distribution

Workato AI Agents - LeadPilot AI and Outlook Automation

- Built LeadPilot AI, a Workato Genie that scores HubSpot leads (0-100) against Salesforce matching context on a five-criterion rubric and maps scores to High (75-100), Medium (45-74), or Low (0-44) priority
- Defined tier-based recommendations (action, owner role, follow-up task) and match-aware handling: exact email matches are treated as existing contacts, company-only matches as distinct people, preventing false duplicates
- Enforced a strict six-field structured output schema for Workato to execute, with guardrails against fabricated facts, unsupported buying-intent inference, and prompt injection; advisory-only, with no CRM write access
- Built an Outlook automation agent (Workato Genie) that summarizes mail, sends new emails and replies, and recommends next actions, orchestrating Workato recipes as tools (get unread, get all, get details, mark as read, send, delete)
- Added a human-approval gate before email deletion

TECHNICAL SKILLS

- **Programming Languages**: Python
- **Generative AI**: LLMs, Agentic AI, AI Agents, Multi-Agent Systems, Prompt Engineering, RAG, Hybrid Search, NLP
- **Agent & LLM Frameworks**: LangGraph, LangChain, Semantic Kernel, Google ADK, Hugging Face
- **Reliability & Debugging**: Validation, Error Handling, Retry and Failover, Logging, Monitoring, Root-Cause Analysis
- **Vector Search**: FAISS, BM25, Cross-Encoder Reranking, Vector Databases
- **Cloud**: Azure OpenAI, Azure AI Foundry Agents, Azure Functions, Azure Identity/AD, Application Insights, Databricks
- **Backend and Data**: FastAPI, REST APIs, PostgreSQL, Cosmos DB
- **Automation**: Workato, Playwright, GitHub Actions (CI/CD)
- **Tools and Platforms**: Git, GitHub, GitLab, Docker, Postman, VS Code, GitHub Copilot, Cursor

EDUCATION

Jawaharlal Nehru Technological University Anantapur (JNTUA) **2019 - 2023**
Bachelor of Technology, Electronics and Communication Engineering

- **Percentage:** 74%